Date: / /

Q1 Inputs:-

M = Master switch (1 = activated)

T = Temperature sensor (1 = $\geq 30°C$)

P = Pressure sensor (1 = ≥ 50 psi)

Logic in Plain English:-

- if M = 1 → light is on, no matter what T and P are
- if M = 0 → light is on only if T = 1 AND P = 1.

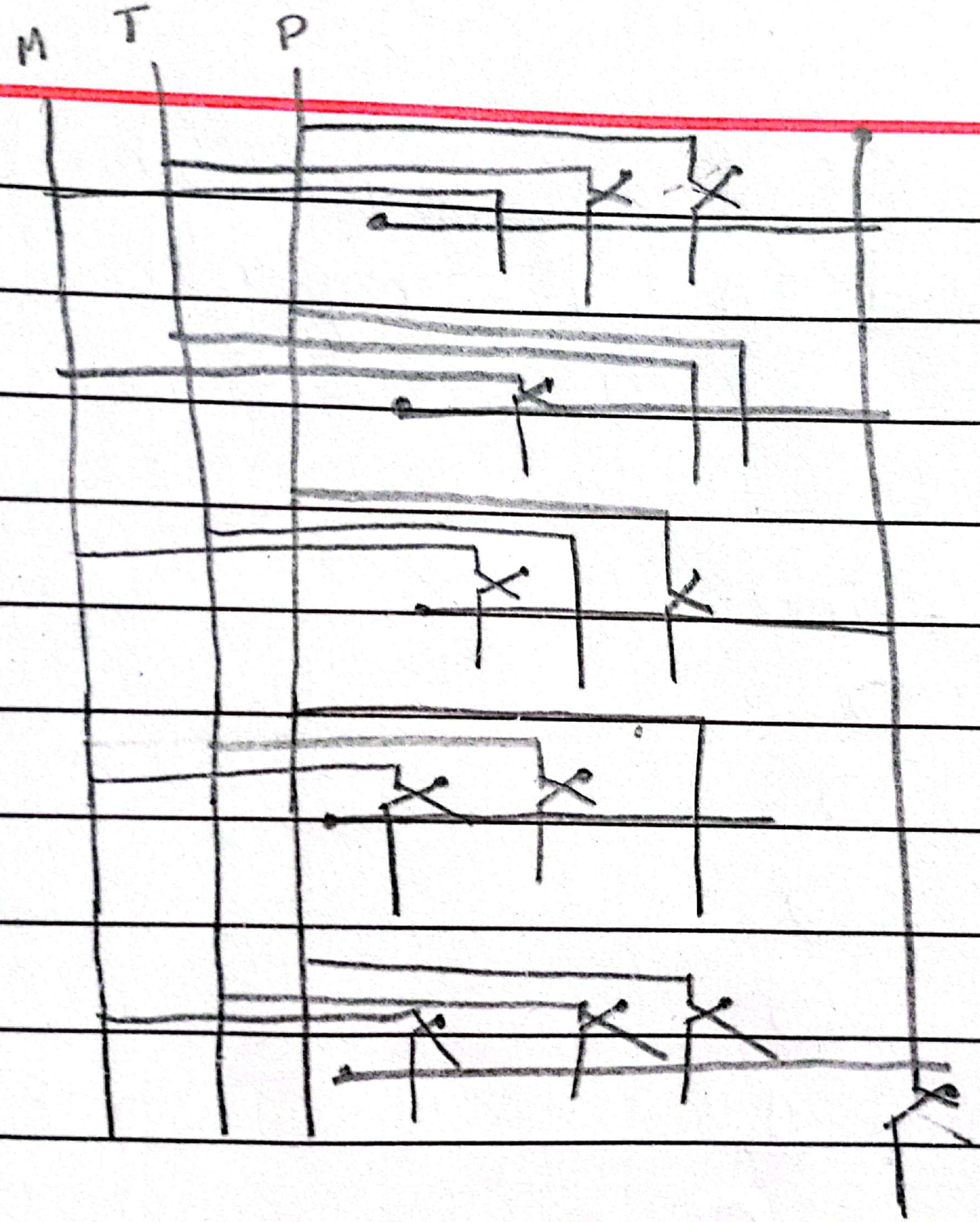
Task A:- Truth Table

M	T	P	Light
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Task B: Boolean Expression

Light = $M + (T \cdot P)$

Task C:-


M
T
P

Date: / /

Inputs:-

Q2) W = water level sensor (1 if > 10cm)

R = Rain expected (1 if yes)

B = Backup battery (1 if charged)

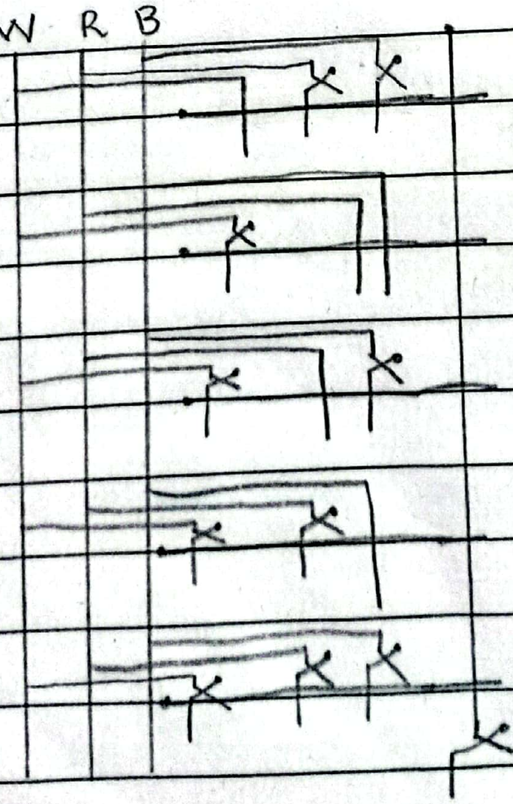
Output = Pump active.

Task A: Truth Table

W	R	B	Pump (P)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Task B: $P = W + (R \cdot B)$

Task C:- W R B



Q3) Inputs:-

T = Primary thermostat (1 if server temperature is > 80°C, else 0)

A = Ambient room sensor (1 if the room temperature is > 25°C, else 0)

M = Maintenance switch (1 if technician is working else 0)

Output F = Cooling ~~Peak~~ fan

Rules:-

1) The ~~if the~~ fan turns on if the server is hot (T=1) and maintanance is not active (M = 0)

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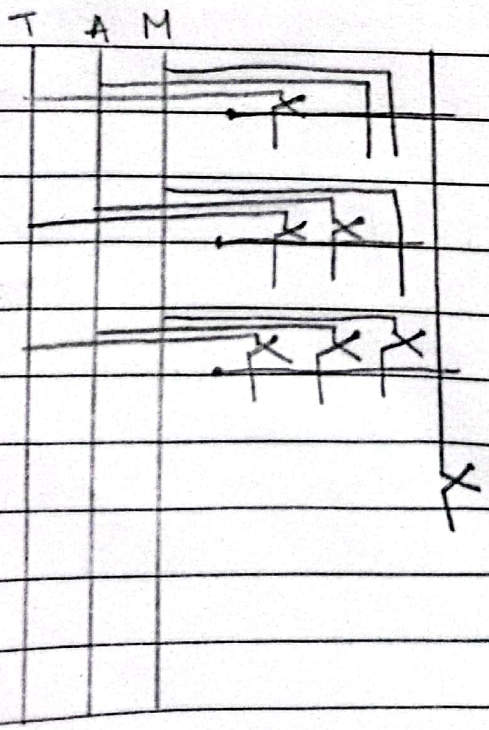
2) The fan also turns on if the server is hot (T=1) and the room is hot (A=1), regardless of maintenance

Task A:

T	A	M	F (Fan)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Task B: $F = T \cdot M' + T \cdot A$



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Q4) Inputs:

M = Manager's key (1 if inserted)

C = Cashier's key (1 if inserted)

T = Time Lock (1 if active between 9 am and 3 pm)

Output : L = Vault Lock opens (1 if open)

The Rules:

- The vault opens only if the Time Lock is active (T = 1) AND atleast one of the two keys (M = 1 or C = 1) is inserted.
- if T = 0, it strictly remains locked

Task A : Truth table

T	M	C	L (vault opens)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Task B : Boolean Expression

$$L = T \cdot M' \cdot C + T \cdot M \cdot C' + T \cdot M \cdot C$$

$$L = T \cdot M' \cdot C + T \cdot M \cdot (C' + C)$$

$$L = T \cdot M' \cdot C + T \cdot M$$

$$L = T \cdot (M' \cdot C + M)$$

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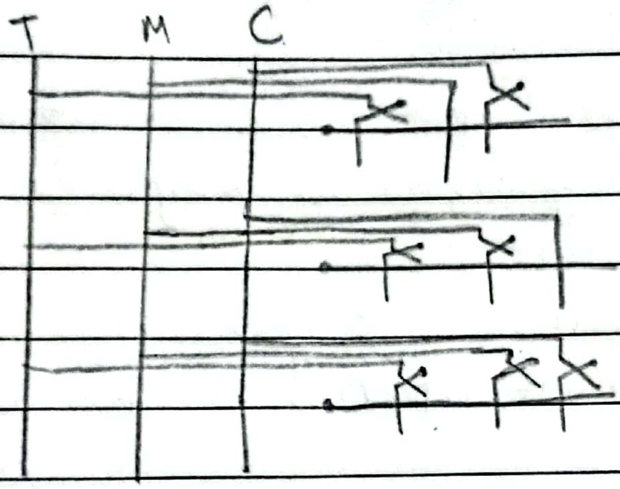
Redundancy Role $M + M'C = M + C$

$L = T\cdot(M+C)$

$L = T\cdot M + T\cdot C$

~~$L = T\cdot(M+C)$~~

Task C: ~~$L = T\cdot M$~~



Q5) Inputs:

T = Exterior Temperature sensor (1 if < 4°C, 0 otherwise)

H = Cabin Humidity sensor (1 if > 70%, 0 otherwise)

D = Manual Defrost botton (1 if pressed)

Output = F = Defroster active (1 is active, 0 is off)

The Rules:

- The defroster activates automatically if the outside temperature is cold and humidity is high (T = 1 and H = 1)
- It also activates any time the manual button is pressed (D = 1), overriding every thing else.

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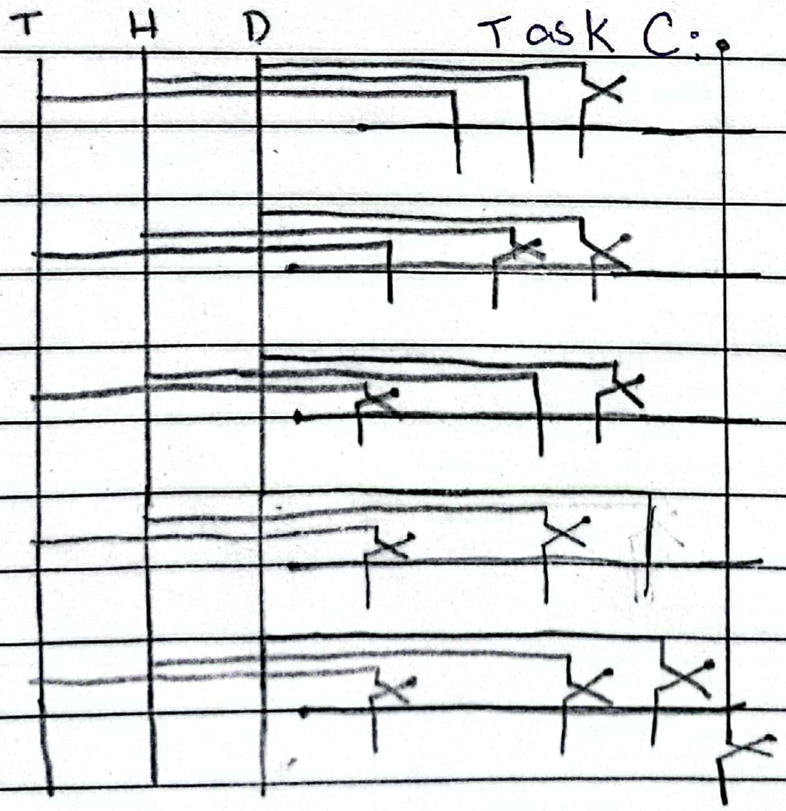
Task A : The Truth Table.

T	H	D	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Task B : Logic Expresion

$$F = T \cdot H + D$$

Task C :



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Q6 Inputs:

W= Weight Scale (1 if > 5kg, 0 if < 5kg

D= Dimension Scanner (1 if too tall > 50cm, 0 if normal weight)

M= Metal Detector (1 if foreign metal is detected, 0 if clean)

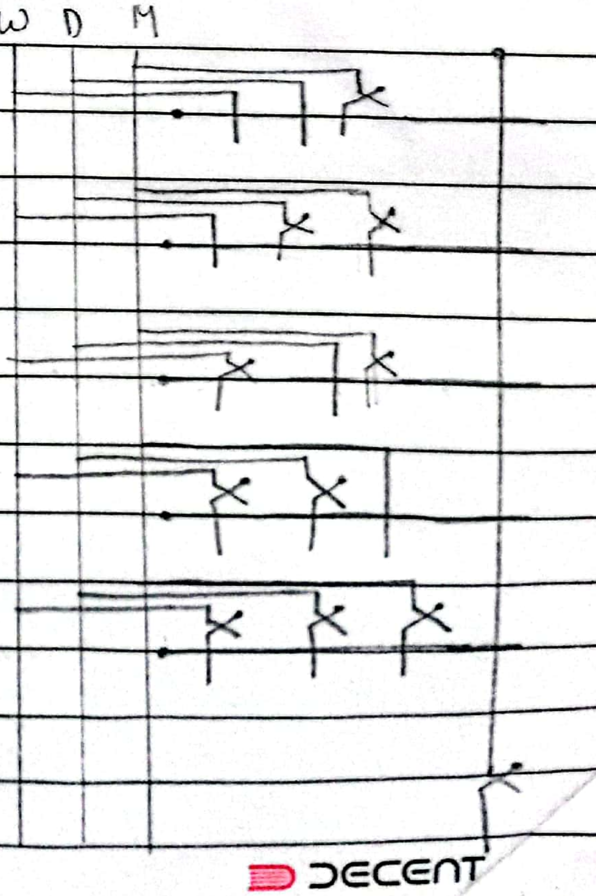
Output R= Reject pneumatic arm (1 if active/ rejects, 0 if lets pass)

Rules:

1) The arm activates immediately if metal is detected (M=1), regarles of size or weight
2) The arm also activates if the package is both overweight (W=1) and too tall (D=1)

Task A: Truth Table

W	D	M	R
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



Task B: Logic Expressions

$R = M + (W \cdot D)$

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Q7) Task A: The Truth Table

Inputs:

B = Mechanical Bumper (1 if it strikes an object, 0 otherwise)

L = Laser Tripwire (1 if broken / person in doorway, 0 otherwise)

W = Weight Limit (1 if > 1000 kg, 0 otherwise)

Output: R = Door Reopens (1 if it reopens, 0 if it continues closing)

Rules:-

1) Reopen immediately if the bumper strikes an object (B = 1)

2) Reopen immediately if the laser tripwire is broken (L = 1)

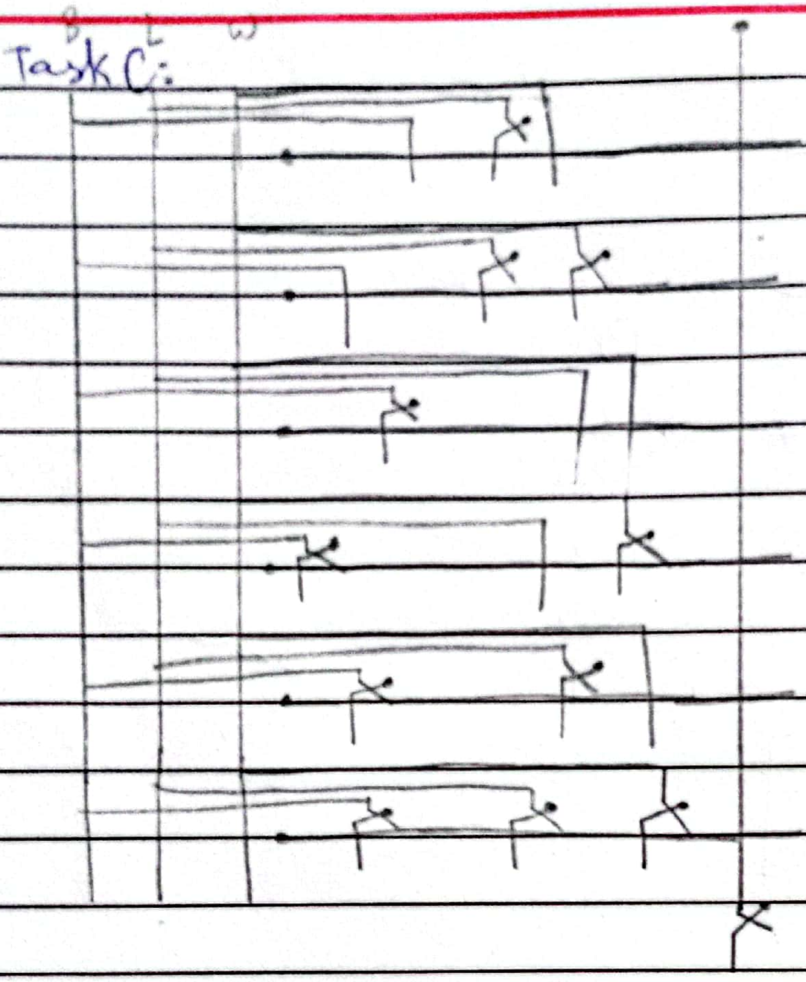
3) Reopen if the weight limit is exceeded and the tripwire is broken (W = 1, L = 1)

B	L	W	R
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Task 2: $R = (B + L) + (W \cdot L)$

Date: / /

Task C:



Q8 Inputs:

S smoke detector (1 if smoke opacity > 20%, 0 otherwise)

H - Heat Sensor (1 if temperature > 75°C, 0 otherwise)

P - Manual Pull Station (1 if pulled, 0 if idle)

~~Relays~~ Output: Y = Fire Sprinkler Trigger (1 if active, 0 if idle)

Rules:-

1) Trigger automatically if both smoke and heat are active.

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Trigger immediately if the manual pull station is activated (P = 1)

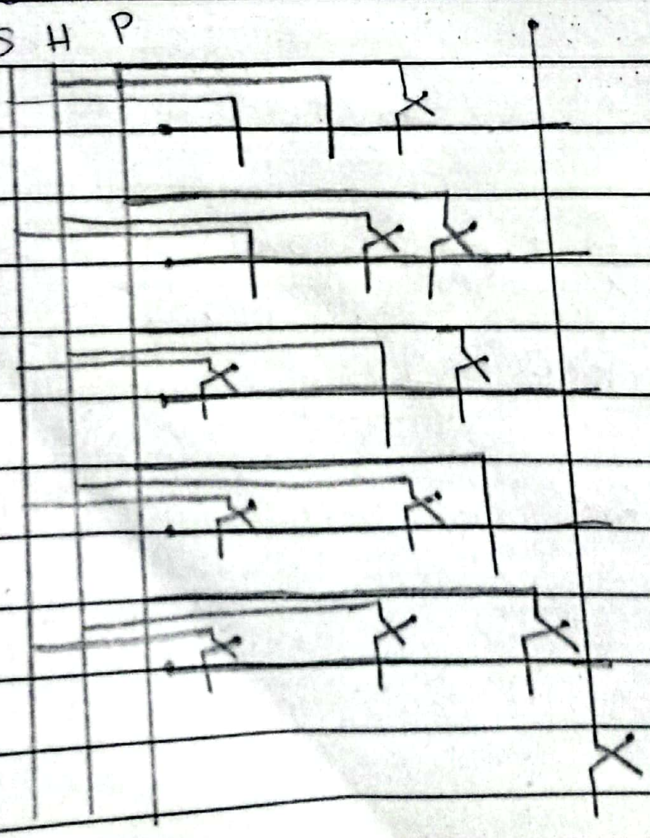
Task A :-

S	H	P	Y (Sprinkler)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Task B:

$Y = (S \cdot H) + P$

Task C: S H P



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Q9) Inputs:

L = Light Sensor (1 if day time >500 lux) 0 if nightime <500 lux)

M = Motion sensor (1 if all chickens are safely inside, 0 if any are outside)

R = Rain Sensor (1 if raining, 0 if dry)

Output: C = Coop door closer (1 closed, 0 if open)

Rules:

1) Close the door if its is nightime (L=0) and all chickens are inside (M=1)

2) Close the door if it is day time (L=1) and raining (R=1) and all chickens are inside (M=1)

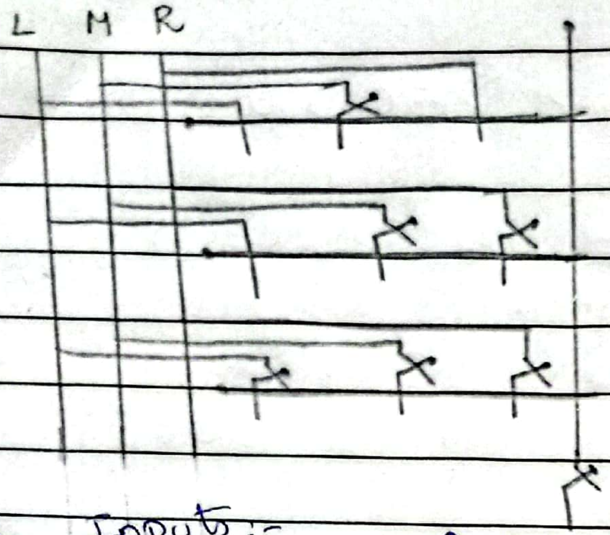
L	M	R	C
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Task B:

$C = (L' \cdot M) + (L \cdot R \cdot M)$

Task C :-

(On Next Page)



Q10) Inputs:
B = Battery level (1 if battery < 15%, 0 otherwise)
D = Distance (1 if distance > 2 km, 0 otherwise)
W = Wind Speed (1 if wind > 20 km/h, 0 otherwise)
Output: A = Battery Save Mode Activates (1 if active, 0 if idle)

Rules:
1) Activate if the battery is low and distance is far (B = 1 . D = 1)
2) Activate if the wind is high and the battery is low (W = 1 . B = 1)

Task B: $A = (B \cdot D) + (B \cdot W)$

Task A:

B	D	W	A
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

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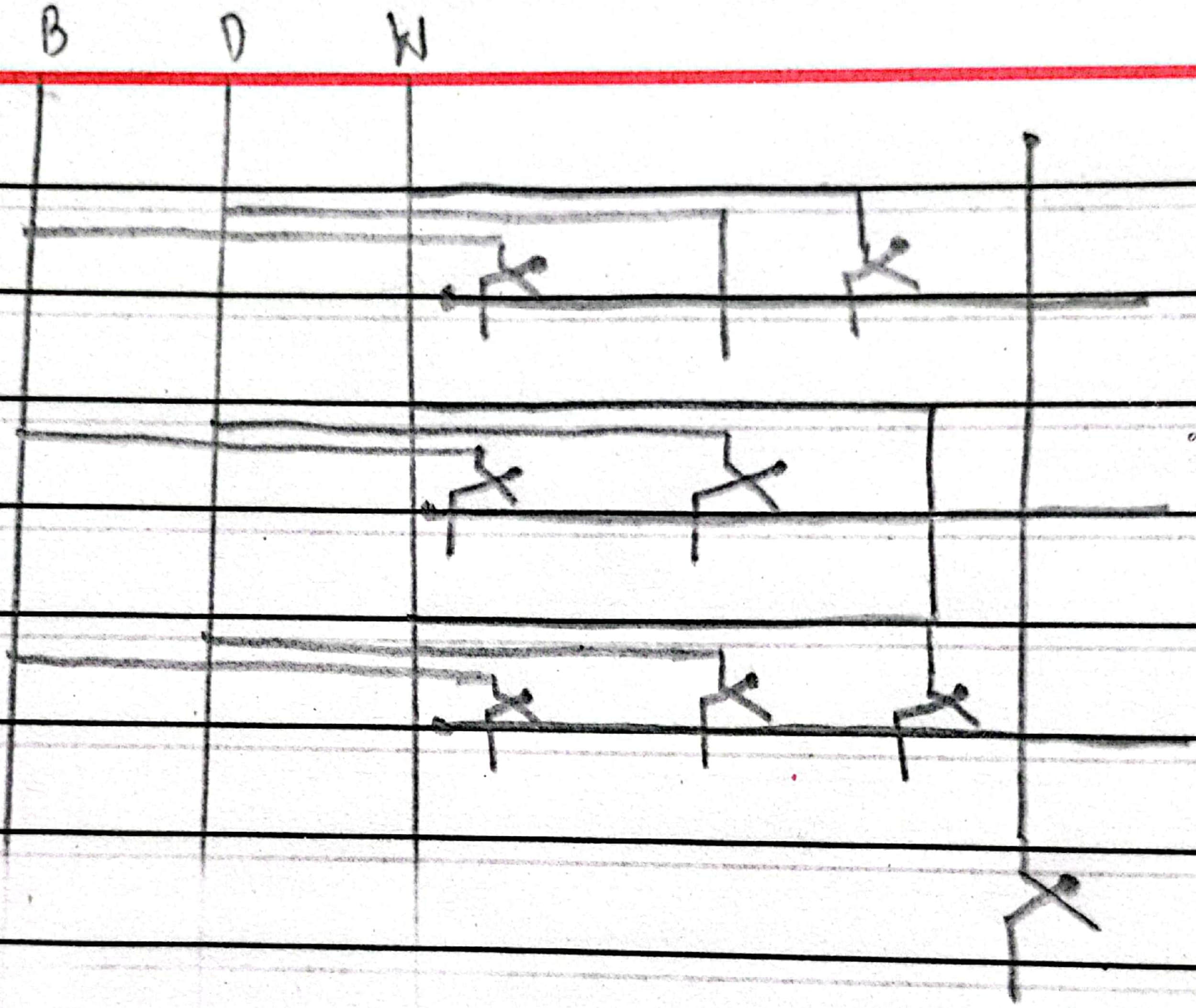

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